Apala Pramanik

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PERSONAL PROFILE

PhD researcher at UNL specializing in ML-driven beam management and power control for mmWave 5G/6G vehicular networks (V2I/V2X). Experienced in SDR-based hardware-in-the-loop testbeds (USRP/GNU Radio), ray-tracing channel modeling (Wireless InSite, Sionna RT), GPU-accelerated deep learning (CUDA/PyTorch), and Integrated Sensing and Communication (ISAC) systems, with expertise in deep reinforcement learning for wireless PHY-layer optimization. Currently expanding into O-RAN architectures and slicing algorithms.

EDUCATION

University of Nebraska–Lincoln – PhD, Computer Science June 2024 – Present
GPA: 3.75/4.0 | Advisor: Dr. Mehmet Can Vuran, Dale M. Jensen Chair Professor *Nebraska, USA*
Dissertation: ML-Driven Beam Management and Power Control for mmWave Vehicular Networks

University of Nebraska–Lincoln – Master's, Computer Science Aug 2021 – May 2024
GPA: 3.75/4.0 | Advisor: Dr. Dung Hoang Tran, Assistant Professor *Nebraska, USA*

Indraprastha University – B.Tech, Electronics and Communication Aug 2017 – May 2021
GPA: 7.9/10 | Thesis: Deep learning techniques for MRI denoising and reconstruction from under sampled MRI scans
Delhi, India

RESEARCH EXPERIENCE

University of Nebraska–Lincoln Nebraska, USA
Graduate Research Assistant – PhD *Aug 2024 – Present*

- Lateral Wave Propagation at mmWave frequencies (IEEE MASS 2026, under review): First experimental demonstration and path loss modeling of lateral waves at mmWave frequencies along a drywall interface, establishing walls as waveguides for indoor coverage
- Camera-Primed Double-Directional Beam Management (IEEE SECON 2026, accepted): Developed a vision-assisted mmWave beamforming framework enabling real-time TX/RX beam selection using camera-derived spatial cues and adaptive correction under hardware constraints.
- Reinforcement Learning for mmWave Power Control: Designed a hardware-in-the-loop RL framework to optimize transmit power and RF gain parameters under SNR constraints using USRP/GNU Radio testbeds and real-time PHY-layer feedback.
- Utilized ray-tracing tools (Wireless InSite, Sionna RT) to model mmWave propagation environments and validate experimental channel observations.
- Beginning exploration of O-RAN architecture and OpenAirInterface, with an early-stage project applying UE slicing at the MAC scheduler
- **Technical Skills:** GNU Radio, USRP, Python, C++, MATLAB, CUDA/PyTorch, Ubuntu Linux, Git

Graduate Research Assistant – Master's *Aug 2021 – May 2024*

- Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems (ACM/IEEE MEMOCODE 2024).
- Applied Neural Network Verification (NNV) for formal analysis of deep learning models in safety-critical systems.
- Specified and verified temporal properties in cyber-physical systems using Signal Temporal Logic (STL).
- **Technical Skills:** ROS (rospy, roscpp), Gazebo, RVIZ, Python, C++, Ubuntu Linux, Git

PUBLICATIONS

Conference Proceedings

Waves on the Walls: Empirical Characterization of mmWave Lateral Waves for Enhanced Indoor Coverage
Apala Pramanik, *Avhishek Biswas, Sasitharan Balasubramaniam, Christos Argyropoulos, Mehmet C. Vuran*
IEEE International Conference on Mobile Ad Hoc and Smart Systems (MASS), 2026 *(under review)*

Look Once, Beam Twice: Camera-Primed Real-Time Double-Directional mmWave Beam Management for Vehicular Connectivity
Apala Pramanik, ***Avhishek Biswas***, *Eylem Ekici, Mehmet Can Vuran*
IEEE International Conference on Sensing, Communication, and Networking (SECON), 2026

Perception-based Runtime Monitoring and Verification for Human-Robot Construction Systems
Apala Pramanik, *Sung Woo Choi, Yuntao Li, Luan Viet Nguyen, Kyungki Kim, Hoang-Dung Tran*
22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE), 2024

ASTITVA: Assistive Special Tools and Technologies for Inclusion of Visually Challenged
Apala Pramanik, *Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, Rohan Tripathi*
International Conference on Computing, Communication, and Intelligent Systems (ICCCIS), 2021

START: Smart Stick based on TLC Algorithm in IoT Network for Visually Challenged Persons
Rahul Johari, Nitesh Kumar Gaurav, Sapna Chaudhary, ***Apala Pramanik***

Fourth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud), 2020

Workshops

Vision-based Runtime Monitoring for Human-Construction Robot Systems
Apala Pramanik, *Kyungki Kim, Dung Hoang Tran*
IROS 2023 Workshop on Formal Methods Techniques in Robotics Systems: Design and Control, 2023

UNIVERSITY PROJECTS

mmWave Propagation Engineering and Bistatic Radar Characterization Spring 2025

- Investigated lateral wave propagation via near-critical incidence on dielectric surfaces; characterized angle-dependent reflection in indoor mmWave environments.
- Built a bistatic radar measurement framework to study material-dependent scattering with relevance to ISAC system design.
- **Technical Skills:** mmWave Propagation, Bistatic Radar, ISAC, RF Measurement, USRP/SDR, Signal Analysis

Multi-Modal Traffic Pattern Detection Using Radio and Camera Sensing Fall 2024

- Built a real-time traffic classification system using synchronized USRP RF signals and camera images; real- and complex-valued CNNs achieved >80% accuracy with <100 ms inference latency via GPU-accelerated execution (CUDA/PyTorch).
- Integrated ±10 ms timestamp synchronization and watchdog mechanism for reliable, continuous operation.
- **Technical Skills:** Python, PyTorch, CUDA, USRP, Signal Processing, Deep Learning (CNN, CVNN), Multi-Modal Data Fusion

mmWave 3D Ray Tracing Framework (MATLAB) Fall 2024

- Built a MATLAB-based 3D ray tracing simulator for mmWave indoor propagation (24 GHz) with multi-bounce reflections, log-normal shadowing, and link budget computation.
- Developed modular environment modeling with power-gradient visualization and automated SVG/EPS/GIF export (Python).
- **Technical Skills:** MATLAB, Python, Ray Tracing, mmWave Channel Modeling, Link Budget Analysis

LLM, Deep Learning and NLP Projects

- A hands-on repo for building LLM agents from scratch and running model training and inference on a GPU cluster.
- Developed a Spanish news classification system using GRU, LSTM, and BETO Transformer models, achieving 91.80% accuracy with tokenizer optimization and data augmentation.
- Implemented CNN image classifiers on Fashion-MNIST and CIFAR-100; sentiment analysis using RNN and Transformer architectures.
- **Technical Skills:** Python, PyTorch, TensorFlow, Pandas, Deep Learning (CNNs, RNNs, Transformers)

INVOLVEMENT

Peer Reviewer	IEEE Transactions on Wireless Communications (2025, 2026)
Undergrad Research Fair	Presented research to undergraduate Computer Science students at UNL
K–12 Outreach	Demonstrated research activities to K–12 students
Mentorship	Mentored undergraduate researchers on experimental design, data analysis, and research communication

SKILLS

ML / DL	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, CUDA
Wireless / SDR	GNU Radio, USRP, Wireless InSite, Sionna RT, mmWave, 5G/6G, V2X/V2I, Beamforming, OFDM, Channel Estimation, ISAC, O-RAN (early stage)
Languages	Python, C/C++, MATLAB, Bash/Zsh
Robotics / CPS	ROS (rospy, roscpp), Gazebo, RVIZ, Point Cloud Library
Tools	Git, Linux, LaTeX, Arduino, Proteus
Coursework	Internet of Things (A+), Deep Learning (A+), Wireless Comm Networks (A+), Advanced Wireless Comm (A), Hardware Acceleration for ML/CUDA (B+)
Soft Skills	Mentoring, Time Management, Teamwork, Problem-solving, Documentation, Engaging Presentation

INTERESTS

Paper Review	Critically evaluating technical rigor, methodology, and scientific contributions in peer-reviewed research.
Tech Writing	Conveying complex research clearly and accessibly through papers and documentation.

References available upon request.